

What the trends say

India is the world's fourth largest manufacturer in the automotive industry, with a large chunk of the global automotive development taking place here. A significant part of that takes place in the software area.

This is not to say that other areas like hardware and mechanical do not have opportunities to grow and build a career. In fact, domains like vehicle body, chassis, infotainment, electrification, vehicle powertrain, and the latest automated driving trend are just as lucrative and also in demand, especially with the evolution that is taking place in the industry.

Additionally, the scope for fields like mechatronics and automation is also growing steadily with the advent and rising popularity of autonomous or driverless mobility. Autonomous driving brings complex systems inside vehicles, and this is accompanied by the need for automation in their development, testing, and even in their manufacture. The scope is huge, and you are learning a mix of mechanical, electronics, and software too. Knowing the software will help you support these technologies.

How to acquire in-demand skills

The automotive sector is growing at a tremendous pace with new innovations, especially in the field of clean mobility. Being knowledgeable in aspects like cybersecurity, data management, data analytics, automation industry 4.0, cloud technologies, AI and ML has become indispensable, and the relevance and demand of these skills are only going to grow.

The Covid pandemic has taught us many lessons, a significant amongst them is that learning can happen anytime, anywhere. Gone are the days when you had to be enrolled in a class or a formal academic course to learn new skills. The Internet has been our greatest ally in sharing and acquiring knowledge, which has enabled us to not rely only on books. Videos, online courses, blogs are some of the few popular ways that people now upskill or reskill themselves. YouTube videos, for one, are the most accessible and cheapest way to learn a new skill, all

without spending a single penny.

Some of the trendiest skills like AI, cybersecurity, and ML are the easiest to find useful resources for. There are also many great options for system engineers and for functional safety and cybersecurity. Programming languages like C++, C, Python, R, Java are in huge demand in this industry, so their knowledge is always going to give you an edge, no matter where you want to specialise in. The reason for this is that most code development on the embedded side takes place on C. There are, of course, other subsets but the above-mentioned are more in demand.

Social media is another great resource for learning and showing your potential. You can always showcase a personal project you have worked on and ask for suggestions from your network (which must include veterans working in the field) to understand where your project idea can be applied. This way, you will also be putting your talents out there for everyone to see, which increases your chances of being noticed and provided with an opportunity by someone looking for a person for that specific skill set. (In fact, I hired a person two years ago after he posted his project on LinkedIn.) Even if the project is not a product or prototype, a well-executed idea can also help to set yourself apart from the normal crowd.

Development or testing?

A common question that many have when starting their career (including myself) is whether to choose development or testing. People tend to assume that testing is somewhat inferior to development. But let me bust the myth. If you have just graduated from college or university, or are new to the industry, entering development does not mean that it is more valuable than testing, or vice versa.

Testing is just as important as developing a new technology or product. When new and complex technologies like advanced driver-assistance systems or autonomous driving systems are being developed, we first test them for their use cases and then further develop them to meet those use cases. This is because we are testing them rather than creating more requirements in the beginning.

When we develop them, we create a prototype based on certain ideas, testing it for certain use-cases and then trying to fix it. This means that the complexity has gradually moved to the testing side in defining new concepts for testing, methods, and even tools while applying the test-driven development or function-driven development method, which is very common these days. Again, this does not mean that development is any less important.

Skills are equally important for both development and testing. You can choose the path you wish to venture into based on the market demands and trends, but also focus on where you can learn more, where you can apply your skills, and which work will satisfy you. Your skills would be optimal in use that way and will support innovation in an area where you are happy to work.

The automotive industry is changing, but so is the competition to make a breakthrough in it. What worked ten years ago may not have that much relevance today. In that case, your skills, knowledge, and experience are what will determine your fate.

Something I recommend to students a lot is the importance of understanding what you are taught. Learning through understanding is very important. Many students might want to study for an exam a night before just to pass the exam, but if you change your goal from merely passing the exam to actually understanding the core concepts, you will definitely reap the benefits of it later.

Moreover, aspire to learn beyond your syllabus. A lot of concepts are not taught in the conventional school or college coursework, but you should not restrict yourself to just those. Embrace the habit of reading. Read whatever you can that is related to technology and its application in different industries. Know what is trending in the industry and what is in demand.

Also maintain a powerful profile on professional networks like LinkedIn and decent exchanges in any social media platform because it can bring you opportunities. **EFY**

The article is based on the talk 'Career guidance: How to upskill and be ready for the automotive revolution' by Plato Pathrose, Advisory Board Member, Vantix/Vinfast, which was presented at MOVES 2021, Tech World Congress 2021